

SplitGuard-AD: A Pre-Training Leakage-Audit Gate for AD MRI Benchmarks

1. Raw 2D MRI slices



Subject A · 5 slices



Subject B · 5 slices

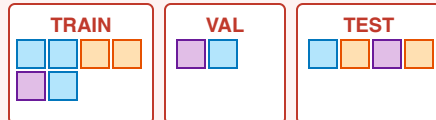


Subject C · 5 slices

(Random)

(Audited)

2a. Random slice split ✗

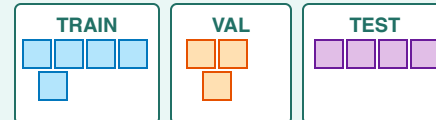


⚠ All 3 subjects appear in train AND test partitions

ADNI1 (N = 382 subjects):
AUROC = 0.949

(inflated — reads training subjects)

2b. Component-safe split ✓



✓ Each subject is fully contained in one partition

ADNI1 (N = 382 subjects):
AUROC = 0.819

(honest — generalises to new patients)

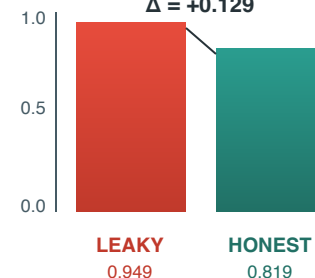
SplitGuard-AD Framework

1. Build leakage graph
subject · session · near-dup edges
2. Freeze component-safe split
no component crosses partitions
3. Pre-training GO / NO-GO audit
PBKDF2-hashed manifest integrity

3. What leakage hides

AUROC comparison

$\Delta = +0.129$



Clinical translation

3.5×

understatement of
missed-diagnosis volume
at screening operating point

Mechanism identified

subject-identity probe:
~211× above chance

Replicated across:

Cohorts — ADNI1 · OASIS-1 · Kaggle
Backbones — ResNet-18 · DenseNet-121

Preserved in every paired-seed bootstrap resample

Direction (leaky > honest) 100% consistent · CIs disjoint from zero
98.3% of gap attributed to inferred subject-identity leakage

Open code + Tier-3 DUA-safe release

github.com/mehmetberkeisler/splitguard-ad
PBKDF2-hashed manifests · Apache-2.0